

Axel **Faes**

Postdoctoral Researcher in AI for Healthcare · Interpretable Tensor Models and Federated Learning for Clinical Prediction

axel.faes[at]utwente.nl | theaxec.github.io | TheAxeC | axelfaes | 0000-0002-1637-255X | Scholar

Research Line

I develop interpretable tensor models and privacy-preserving federated learning for clinical prediction from biomedical signals and imaging, so that clinicians can read the models and the models can move between hospitals without the patient data.

- **Interpretable block-term tensor models**: published [5, 7-8, 13-15], in review [20-22, 24-27, 29]
- **Federated and privacy-preserving learning**: published [4, 11, 14-15], in review [19, 26-27, 29]
- **Clinical use on ECG, liver MRI and patient outcomes**: published [1, 11-12], in review [22-23, 28, 30]

Work & Research Experience

Postdoctoral Researcher: Foundation Models & Multi-Modal Federated Learning for Precision Medicine and Liver Cancer Detection

University of Twente / Medisch Spectrum Twente
Jan. 2026 - Current

- Developing advanced deep learning and foundation models integrating longitudinal MRI data, multi-omics profiles, and clinical variables for early detection and risk prediction of hepatocellular carcinoma (HCC).
- Building explainable AI (XAI) frameworks that link imaging features and molecular markers to clinically meaningful outcomes in precision oncology within the AI-HCC ZonMw project, in the CoDE group (Cognition, Data and Education).

Postdoctoral Researcher: Federated Learning in Healthcare & Real-World Evidence

UHasselt
Jan. 2024 - Jan. 2026

- Scientific Coordinator of the Flanders AI Research Program Use Case Real World Evidence, leading multi-institutional research on privacy-preserving healthcare analytics. As Technical Machine Learning Lead of the Biomedical Data Sciences Group (Prof. Liesbet Peeters), developed federated learning frameworks enabling collaborative clinical research across Belgian healthcare institutions without sharing sensitive patient data.
- Research encompasses federated tensor regression for longitudinal health data, distributed learning architectures for heterogeneous clinical datasets, and translation of federated AI methods into practical healthcare applications. Research Affiliate from 2026.

Postdoctoral Researcher: Deep Learning & Tensor Regression for Sign Language Alphabet Decoding via Brain-Computer Interfaces

KU Leuven
May 2023 - Jan. 2024

- Developed advanced machine learning methods for decoding sign language alphabet gestures from high-density intracranial brain activity (ECoG). Combined deep learning architectures with tensor regression models to capture complex spatiotemporal patterns in neural signals associated with linguistic motor control.
- Technical contributions included implementing block-term tensor decomposition for multi-dimensional neural data and exploring cross-subject generalization for robust BCI performance. The research advances both fundamental understanding of neural encoding of sign language and practical assistive communication technology. Conducted at the Laboratory for Neuro- and Psychophysiology under supervision of Prof. Marc Van Hulle, extending previous doctoral research in motor decoding to linguistic applications.

Data Science Consultant

Flanders BCI
2022 - 2024

- Provided consultancy for physical experiments using Synamps RT and Neuvo EEG Systems.
- Supported the data analysis for hybrid BCI with new analytical EEG modeling techniques & signal detection- and classification algorithms.
- Infer emotion from EEG signals using machine learning techniques.
- Usage of ECoG signals from semi-invasive recording for imagined movement detection and language processing.

Research Stay Idrogenet Srl, Gloreha: Robotic Rehabilitation

Lumezzane, Italy
Dec. 2022 - Jan. 2023

- Develop custom software for the control of the Gloreha robotic rehabilitation device,
- Specifically the R-TOUCH PRO Sinfonia, a custom-built robotic exoskeleton for hand rehabilitation.

PhD Researcher in Computational Neuroscience

KU Leuven
Sep. 2018 - May 2023

- Cognitive and Molecular Neuroscience
- PhD Thesis: Finger Movement Decoding: From Source-Localisation to Tensor Regression Modelling
- Supervisor: Prof. Marc Van Hulle

Web Performance Research Internship

UHasselt
Jul. 2016 - Sep. 2016

- I worked on the iMinds PRO-FLOW project @ Expertise centre for Digital Media (EDM)
- focus on the difference between the http versions (http1.1, https, http2).

Summer Research Internship Physical Computing

UHasselt
Aug. 2015 - Sep. 2015

- Interfacing between human entity, a drone and virtual objects @ Expertise centre for Digital Media (EDM).
- (C++, Optitrack motion capture, custom built drone)

Teaching & Academic Activities

Bioinformatics (3740) Coordinating lecturer, UHasselt — *Sep. 2025 - Sep. 2026*
BKO (Basic Teaching Qualification), Teacher Professionalization, completed, UHasselt — *2025*
Data Science in Healthcare (4747) Lecturer (teaching team member), UHasselt — *Sep. 2025 - Dec. 2025*
Advanced Topics in Data Science (4569 & 4585) Lecturer Federated Learning (teaching team member), UHasselt — *Jun. 2024 - Sep. 2025*
Bioinformatics (3740) Coordinating lecturer, UHasselt — *Sep. 2024 - Sep. 2025*
Brain-Computer Interfacing (B-KUL-H08M0A) Guest Lecturer, KU Leuven — *2022 - 2025*
MSc thesis assessor (second reader), UHasselt — *2026*
Data Science Support for PhD Students (BIOMED), data science expert, UHasselt — *2024*

Academic Service & Peer Review

Guest Associate Editor Use of Big Data and Artificial Intelligence in Multiple Sclerosis, Frontiers in Immunology — *2024 - 2025*

Journal peer reviewer, npj Digital Medicine, Bioinformatics (OUP), ACM Computing Surveys, IEEE TNNLS, Journal of Neural Engineering, Neurocomputing, International Journal of Neural Systems, Computational Biology and Chemistry, Frontiers in Neuroscience, MDPI journals (Algorithms, Applied Sciences, Axioms, Bioengineering, Brain Sciences, Electronics, Future Internet, Robotics, Signals)

Conference peer reviewer, AAAI 2024, FLTA 2026, FLLM 2026, CBI & EDOC 2026 — *2024 - 2026*

VLAIO Evaluator for BAEKELAND SPIN OFF MANDATES, VLAIO — *2025*

Student Research Competition Judge (International Conference on Functional Programming), ICFP — *2024*

Educational background

Doctoral Programme in Biomedical Science (PhD) in Computational Neuroscience
KU Leuven — *Sep. 2018 - May 2023*

- Cognitive and Molecular Neuroscience
- PhD Thesis: Finger Movement Decoding: From Source-Localisation to Tensor Regression Modelling

Advanced Master of Science in Engineering (M.Sc.) in Artificial Intelligence
KU Leuven — *Sep. 2017 - Jul. 2019*

- Engineering and Computer Science (76% - Cum Laude)
- Thesis: An Information Theoretical Approach to EEG Source-Reconstructed Connectivity (on Github)

Honours programme of the Faculty of Engineering Science (Research Track)
KU Leuven — *Sep. 2016 - Oct. 2018*

- Research Assistant: design of type-&-effect system for Eff based on row polymorphism
- Research Assistent: efficient compilation of algebraic effect handlers (in Eff)

Master of Science in Engineering (M.Sc.) in Computer Science (Burgerlijk Ingenieur - ir.)
KU Leuven — *Sep. 2016 - Sep. 2018*

- Artificial Intelligence & Theoretical Computer Science (76% - Cum Laude)
- Thesis: Algebraic Subtyping for Algebraic Effects and Handlers (on Github)

Business Summer School: United in Manchester (0739)
The University of Manchester — *Jul. 2015 - Aug. 2015*

- International Business

Bachelor of Science (B.Sc.) in Computer Science
UHasselt — *Sep. 2013 - Jul. 2016*

- Physics and General courses (79% - Magna Cum Laude)
- Thesis: Machine learning techniques for flow-based network intrusion detection systems (on Github)

Flemish Sign Language course
Flemish Sign Language Centre — *Sep. 2022 - May 2024*

- Graduates are able to engage in day-to-day conversations with deaf and hard-of-hearing individuals.

Grants & Funding

The Self-Explaining Hospital: Readable, Private Clinical Models That Travel Between Institutions
Applicant — *2026*

- Status: Submitted, decision pending
- Funder: NWO, Veni scheme (pre-proposal)

Before the tumour: an interpretable pre-lesional field trajectory that anticipates lesion emergence in the cirrhotic liver
Applicant — *2026*

- Status: Submitted, decision pending
- Funder: NWO, Open Competition Domain Science XS (ENW-XS), Life Sciences cluster

The kidney allograft biopsy as a measurement: read as a trajectory, corrected between sites, reported with a stated confidence
Applicant — *2026*

- Status: Submission November 2026, decision pending
- Funder: FWO, Senior Postdoctoral Fellowship

Finger movement decoding: from source-localisation to graph-based regression modelling
PhD Fellow — *2018*

- Status: Awarded (FWO 1157019N)
- Funder: FWO, PhD Fellowship (Fundamental Research)

Projects

AI-HCC: Predicting the Development of Primary Liver Cancer Using the AI-HCC Model - ZonMw
Postdoctoral Researcher — *2026 - Current*

- Role: Postdoctoral researcher (University of Twente / Medisch Spectrum Twente)
- Funder: ZonMw, The Netherlands Organisation for Health Research and Development

Use Case Real World Evidence - Flanders AI Research Program
Scientific Coordinator — *2024 - Jan. 2026*

- Role: Scientific coordinator (UHasselt) and AI Researcher
- Team: approx. 10 researchers across 4 institutions

ELIXIR Belgium - Roadmap 2023-2025
Consortium Partner — *2023 - 2025*

- Role: Consortium partner (UHasselt), focusing on health data re-use and federated analyses
- Collaborations: KU Leuven (Yves Moreau), University of Antwerp (Geert Vandeweyer), VIB (Frederik Coppens)
- Context: ELIXIR Belgium is part of ELIXIR Europe, an ESFRI initiative funded by the European Commission and national agencies

Federated Learning Kit (FLkit)

Contributor
2025 - Current

- FLkit is designed to help life scientists apply federated learning in their research. It offers guidance, techniques, and tools for working with decentralized and sensitive data, enabling privacy-preserving collaboration and deeper insights without sharing raw data.
- The project is in its early stages and open to contributors. The long-term goal is to make FLkit community-driven, sustained, and governed by researchers and practitioners who want to advance federated learning for life sciences.

Cardinal: Educational and Research platform for MSc theses in PL/compilers

Lead Developer
Jan. 2015 - Current

- Cardinal is a C++20 reimplementation of the Wren VM, designed for clarity, research, and education. It provides a small, well-documented virtual machine and runtime for exploring compilers, concurrency (fibers), and language embedding.
- The project serves as a platform for student theses and for prototyping new programming language ideas.

Open Source Python Package - Block-Term Tensor Regression (BTTR)

Lead Researcher & Developer
2022 - Current

- The open-source BTTR implementation provides researchers and practitioners with accessible, reproducible tools for tensor-based regression in neuroscience and related domains. It is actively developed and documented to support experimentation, education, and further research.

Wisconsin Breast Cancer Dataset Research Tutorial for Federated Learning

Research Tutorial Author
2025 - Current

- This repository serves as a step-by-step tutorial for running federated learning studies with the FL4E framework. Using the Wisconsin Breast Cancer dataset as an example, it guides researchers and students through the full workflow.
- Clear documentation and markdown guides explain each step, making it easier for newcomers to understand and apply federated learning in practice.

Oculus

Lead Developer
2026 - Current

- An open-source desktop workbench that brings the projects, manuscripts, grants, data and code of a research hub into one VS Code-like app.

Honors & Awards

Jun. 2025	**Selected for EFSA Framework Contract**, European Food Safety Authority	*Europe*
Mar. 2018	**Finalist**, Cyber Security Challenge 2018	*Brussels, Belgium*
Sep. 2017	**3rd place**, ICFP 2017 Student Research Competition	*Oxford, UK*
Jul. 2016	**Bachelor Award**, in Computer Science	*UHasselt, Belgium*
May. 2016	**3rd place**, ACM CHI 2016 Student Design Competition	*San Jose, CA, USA*
Feb. 2016	**2nd place**, BeGDC (Belgian Game Development Championship)	*Brussels, Belgium*

Professional Development Activities

Twente Health School (summer school), University of Twente, Interdisciplinary health-research summer school at the University of Twente. *2026*

Mathematics & Natural Sciences Tutor, Freelance, Supporting high school and university students in mastering mathematics (and other natural sciences) through personalized tutoring *Sep. 2015 - 2024*

Coach, DjangoGirls, Inspire women to fall in love with programming (Python, Django workshops) *Mar. 2018 - 2024*

Coach, CoderDojo (UHasselt, PXL), Teach children programming (Scratch, Python, Minecraft and Lego mindstorm). *Sep. 2014 - 2025*

Student Representative, KU Leuven, POC (Education Committee) of Master Computer Science Engineering, POC (Education Committee) of Advanced Master Artificial Intelligence, Member of Departmental council of Computer Science, Department board of Computer Science and Faculty council of Engineering Science *Sep. 2017 - Sep. 2018*

IELTS Academic Module (8.0/9.0), Brussels, Belgium, English language qualification. *Jan. 2016*

Publications

International Journal Papers

[1] Dongho Chun, **Axel Faes**, "Which representation clusters? A chance-corrected benchmark of ECG representations for unsupervised arrhythmia structure", Biomedical Signal Processing and Control, 2026.

[2] Liesbet Peeters, **Axel Faes**, et al., "Editorial introduction 'The use of big data and AI in MS'", Frontiers in Immunology, 2025.

[3] Valentina Pergher*, **Axel Faes***, Yide Li, Marc M. Van Hulle (* co-first author), "How stimulus type and task structure affect ERP signatures during an N-Back task", Cognitive Processing, 2026.

[4] Ashkan Pirmani, Edward De Brouwer, Adam Arany, Martijn Oldenhof, Antoine Passemiers, **Axel Faes**, Tomas Kalincik, Serkan Ozakbas, Riadh Gouider, Barbara Willekens, Dana Horakova, Eva Kubala Havrdova, Francesco Patti, Alexandre Prat, Alessandra Lugaresi, Valentina Tomassini, Pierre Grammond, Elisabetta Cartechini, Izanne Roos, Cavit Boz, Raed Alroughani, Maria Pia Amato, Katherine Buzzard, Jeannette Lechner-Scott, Joana Guimaraes, Claudio Solaro, Oliver Gerlach, Aysun Soysal, Jens Kuhle, Jose Luis Sanchez-Menoyo, Daniele Spitaleri, Tunde Csepany, Bart Van Wijmeersch, Radek Ampapa, Julie Prevost, Samia J. Khoury, Vincent van Pesch, Nevin John, Davide Maimone, Bianca Weinstock-Guttman, Guy Laureys, Pamela McCombe, Yolanda Blanco, Ayse Altintas, Abdullah Al-Asmi, Justin Garber, Anneke van der Walt, Helmut Butzkueven, Koen de Gans, Csilla Rozsa, Bruce Taylor, Talal Al-Harbi, Attila Sas, Cecilia Rajda, Orla Gray, Danny Decoo, William M Carroll, Allan G Kermode, Marzena Fabis-Pedrini, Deborah Mason, Angel Perez Sempere, Mihaela Simu, Neil Shuey, Bhim Singhal, Marija Cauchi, Todd A. Hardy, Sudarshini Ramanathan, Patrice Lalive, Carmen-Adella Sirbu, Stella Hughes, Tamara Castillo Trivino, Liesbet M. Peeters, and Yves Moreau, "Personalized Federated Learning for Predicting Disability Progression in Multiple Sclerosis Using Real-World Routine Clinical Data", npj Digital Medicine, 2025.

[5] **Axel Faes**, Eva Calvo Merino, Mariana P. Branco, Anais Van Hoylandt, Elina Keirse, Tom Theys, Nick F. Ramsey, Marc M. Van Hulle, "Decoding Sign Language Finger Flexions from high-density electrocorticography using Graph-Optimized Block Term Tensor Regression", Journal of Neural Engineering, 2025.

[6] Eva Calvo Merino, **Axel Faes**, Marc M. Van Hulle, "The role of distinct ECoG frequency features in decoding finger movement", Journal of Neural Engineering, 2023.

[7] **Axel Faes**, Marc M. Van Hulle, "Finger movement and coactivation predicted from intracranial brain activity using extended Block-Term Tensor Regression", Journal of Neural Engineering, 2022.

[8] **Axel Faes**, Flavio Camarrone, Marc M. Van Hulle, "Single finger trajectory prediction from intracranial brain activity using Block-Term Tensor Regression with fast and automatic component extraction", IEEE Transactions on Neural Networks and Learning Systems, 2022.

[9] **Axel Faes**, Aurelie de Borman, Marc M. Van Hulle, "Source space reduction for eLORETA", Journal of Neural Engineering, 2021.

[10] **Axel Faes**, Iris Vantieghem, Marc M. Van Hulle, "Neural Networks for Directed Connectivity Estimation in Source-Reconstructed EEG Data", Applied Sciences, 2022.

Conference Papers

[11] **Axel Faes**, Maureen M. J. Guichelaar, Donald E. Bouman, Stéphanie M. van den Berg, Maryam Amir Haeri, "Federate What Transfers: Component-Aware Federation of Self-Supervised Backbones for Longitudinal 3D MRI", IEEE Conference on Federated Learning Technologies and Applications (IEEE FLTA 2026).

[12] **Axel Faes**, "Transparent and Robust: Interpretable Inputs Improve Foundation Models in Mission-Critical Decision Pipelines", XAI-EADM Workshop at CBI-EDOC 2026, Springer LNBIP.

[13] **Axel Faes**, Eva Calvo Merino, Anais Van Hoylandt, Elina Keirse, Tom Theys, Marc M. Van Hulle, "Finger abduction trajectory prediction from high-density ECoG", 10th Graz Brain-Computer Interface Conference 2026.

[14] **Axel Faes**, Liesbet Peeters, "Optimizing Federated Block-Term Tensor Regression: Strategy Comparisons and Applications", The 3rd International Conference on Foundation and Large Language Models (FLLM2025), Symposium on Federated Learning and Intelligent Computing Systems (FLICS 2025).

[15] **Axel Faes**, Ashkan Pirmani, Yves Moreau, Liesbet Peeters, "Applying Federated Learning to Block-Term Tensor Regression for Decentralised Data Analysis of Biomedical Data", IEEE Conference on Federated Learning Technologies and Applications (IEEE FLTA 2025).

[16] Qiang Sun, Eva Calvo Merino, Liuyin Yang, **Axel Faes**, Marc Van Hulle, "On the Impact of Proprioception in EEG Representations and Decoding during Human-Hand Exoskeleton Interaction", Proceedings of 2025 International Conference on Rehabilitation Robotics (ICORR) 2025.

[17] Robin Marx, Maarten Wijnants, Peter Quax, **Axel Faes**, Wim Lamotte, "Web Performance Characteristics of HTTP/2 and comparison to HTTP/1.1", International Conference on Web Information Systems and Technologies, pg 87-114, 2018.

[18] Robin Marx, Peter Quax, **Axel Faes**, Wim Lamotte, "Concatenation, embedding and sharding: Do HTTP/1 performance best practices make sense in HTTP/2?", WEBIST 2017 - Proceedings of the 13th International Conference on Web Information Systems and Technologies.

Papers in Review

[19] Ashkan Pirmani*, Ilse Vermeulen*, Goran Vinterhalter, Lotte Geys, **Axel Faes**, Muhammad Quamber Ali, Nishkala Sattanathan, Geert Vandeweyer, Yves Moreau, Liesbet M. Peeters (* co-first author), "Development and Design of FLKit: A Structured Onboarding Toolkit for Federated Learning in Health and Life Sciences", Journal of Medical Internet Research (JMIR), 2026.

[20] Ward Ceyssens, **Axel Faes**, "Behavior-Supervised Alignment for Cross-Subject Neural Decoding across Non-Corresponding Electrode Grids", Neural Networks, 2026.

[21] **Axel Faes**, Maryam Amir Haeri, "Registration-Free Supervised Tensor-PLS for Unaligned Multiparametric MRI", Pattern Recognition, 2026.

[22] **Axel Faes**, Liesbet Peeters, "An Interpretable Block-Term Tensor Network for Incident Atrial-Fibrillation Prediction from the Sinus-Rhythm 12-Lead ECG and Clinical Risk Factors", Artificial Intelligence in Medicine, 2026.

[23] Meseret Assefa Kerga, **Axel Faes**, "Competing-Risks Modelling of a Widely-Used Cirrhosis Outcome Benchmark: Recovering the Transplant Outcome and Cause-Specific Risk Lost to Classification", Computers in Biology and Medicine, 2026.

[24] Dries Cornelissen, **Axel Faes**, "An interpretable tensor model for identifying atrial fibrillation in sinus-rhythm ECG: block-term tensor regression on PTB-XL, benchmarked against deep and foundation models", Medical & Biological Engineering & Computing, 2026.

[25] **Axel Faes**, Maryam Amir Haeri, "Jointly localized spatiotemporal components of evoked EEG via a lead-field-constrained block-term decomposition", Journal of Neuroscience Methods, 2026.

[26] **Axel Faes**, Maryam Amir Haeri, "Privacy-preserving federated tensor decomposition of single-cell immune data: recovering multicellular programs across institutions", Computer Methods and Programs in Biomedicine, 2026.

[27] **Axel Faes**, Maryam Amir Haeri, "Private Coupled Descent: Dimension-Free Privacy for Vertical Federated Learning by Privatizing the Coupling Variable", IEEE Transactions on Signal Processing, 2026.

[28] **Axel Faes**, "You Don't Need the Black Box: Named ECG Features Rival Foundation Models for Mortality Prediction", Computers in Biology and Medicine, 2026.

[29] **Axel Faes**, Maureen M. J. Guichelaar, Donald E. Bouman, Maryam Amir Haeri, "When Masking Meets Sparsity: COVERT for Private, Communication-Efficient Vertical Federated Tensor Regression", Neurocomputing, 2026.

[30] Jakob Nolte*, **Axel Faes***, Maureen M. J. Guichelaar, Donald E. Bouman, Stéphanie M. van den Berg, Maryam Amir Haeri (* co-first author), "HCCNet: A Self-Supervised CNN-Transformer for Disease Prognosis from Longitudinal 3D MRI, with a Case Study on Hepatocellular Carcinoma", Intelligent Medicine, 2026.

Papers in Preparation

[31] **Axel Faes***, Jasper van Leengoed*, Maryam Amir Haeri, Donald Bouman, Stephanie van den Berg, Maureen Guichelaar (* co-first author), "Prediction of hepatocellular carcinoma three years ahead from surveillance MRI of the liver parenchyma", In preparation.

[32] **Axel Faes**, Maryam Amir Haeri, "A Generalization Separation for Block-Term Spectral Operators", In preparation.

[33] **Axel Faes**, Maryam Amir Haeri, "MoBTE: A Mixture of Block-Term Spectral Experts for Multi-PDE Operator Learning", In preparation.

Other publication

[34] Matija Pretnar, Amr Hany Shehata Saleh, **Axel Faes**, Tom Schrijvers, "Efficient compilation of algebraic effects and handlers", CW Reports, CW708, 32 pp. Leuven, Belgium: Department of Computer Science, KU Leuven, 2017.

Thesis

[35] **Axel Faes**, "Finger Movement Decoding: From Source-Localisation to Tensor Regression Modelling", PhD Thesis, 2023.

Extended Abstracts

[36] Eva Calvo Merino, **Axel Faes**, Marc M. Van Hulle, "High-gamma band event detection improves stability of finger trajectories decoded from ECoG-LMP activity", International BCI Meeting 2024.

[37] Qiang Sun, **Axel Faes**, Marc M. Van Hulle, "Individual and Coordinated Finger Movements Decoding from High-Density EEG and Its Implication in Hand Exoskeleton Control", European Congress of NeuroRehabilitation 2023.

[38] Eva Calvo Merino, **Axel Faes**, Marc M. Van Hulle, "Modulation of LMPs using the gamma band increases the stability of finger trajectories decoded from ECoG", BCI Society Meeting 2023.

[39] **Axel Faes**, Benjamin Wittevrongel, Marc M. Van Hulle, "Reconstructing single finger trajectories from intracranial brain activity", III International Conference "Volga Neuroscience Meeting 2021".

[40] **Axel Faes**, Mansoureh Fahimi Hnazaee, Marc M. Van Hulle, "Causal Graphical Modelling of Functional Connectivity from Reconstructed EEG Sources", 8th International BCI Meeting (2021).

[41] **Axel Faes**, Tom Schrijvers, "Towards a Core Language with Row-Based Effects for Optimised Compilation", International Conference on Functional Programming 2017 Student Research Competition.

[42] Kashyap Todi, Donald Degraen, Brent Berghmans, **Axel Faes**, Matthijs Kaminski, Kris Luyten, "Purpose-centric appropriation of everyday objects as game controllers.", CHI EA '16: Extended Abstracts of the SIGCHI Conference on Human Factors in Computing Systems. Student Game Competition, pp. 2744-2750, 2016.

[43] Brent Berghmans*, **Axel Faes***, Matthijs Kaminski*, Kashyap Todi (* co-first author), "Household Survival: Immersive Room-Sized Gaming Using Everyday Objects as Weapons", CHI EA '16: Extended Abstracts of the SIGCHI Conference on Human Factors in Computing Systems. Student Game Competition, pp. 168-171, 2016.

Posters

[44] **Axel Faes**, Maureen M. J. Guichelaar, Donald E. Bouman, Stéphanie M. van den Berg, Maryam Amir Haeri, "Modality-Specific Encoding with a Shared Backbone for Early HCC Prediction from Multi-Modal Liver MRI", XAI-NL Meet-up, 2026.

[45] **Axel Faes**, Tom Schrijvers, "Towards a Core Language with Row-Based Effects for Optimised Compilation", International Conference on Functional Programming 2017 Student Research Competition.

Supervised students, talks and other media

Students

[1] **Brecht Heeren**, "Federated Machine Learning for Health Data", Master of Science in Computer Science (2025 - 2026).

[2] **Denzell Mgbokwere**, "Optimization and Type Checking in Single-Pass Compilers - a Case Study with the Wren Programming Language", Master of Science in Computer Science (2025 - 2026).

[3] **Robert Rysskin**, "Redesign of Wren Bytecode - Towards More Efficient Execution and Memory Usage", Master of Science in Computer Science (2025 - 2026).

[4] **Anh Phuong DO**, "Individual Reference Intervals for Clinical Event Prediction", Doctoral Program in Sciences, Statistics, co-supervisor (2024 - 2025).

[5] **Matteo Ramina**, "Estimating Household Wealth in Guyana - Remote Sensing and Convolutional Neural Network Approach", Master Statistics & Data Science (2024 - 2025).

[6] **Mohsen Soleimanisemsani**, "AI for time series imputation", Master Statistics & Data Science (2024 - 2025).

[7] **Meseret Assefa Kerga**, "Predicting Cirrhosis Patient Survival Using Machine Learning - A Data-Driven Approach", Master Statistics & Data Science (2024 - 2025).

[8] **Dongho Chun**, "Clustering with Cardiovascular Health Data", Master of Science in Computer Science (2024 - 2025).

[9] **Dries Cornelissen**, "BTTR for arrhythmia detection/prediction on sinus rhythms", Master of Science in Computer Science (2024 - 2025).

[10] **Ward Ceyssens**, "Cross Subject training for finger movement decoding with high-density ECoG", Master of Science in Computer Science (2024 - 2025).

[11] **Qiang Sun**, "Hand exoskeleton dexterity achieved by shared control with a semi-invasive brain-computer interface", Doctoral Program in Biomedical Sciences, daily supervision (2022 - 2023).

[12] **Eva Calvo Merino**, "Restoring finger dexterity with an exoskeleton controlled by human intracranial recordings", Doctoral Program in Biomedical Sciences, daily supervision (2022 - 2023).

[13] **Aurélie de Borman**, "Investigating the effect of Source Mixing on Directed Connectivity estimated between Simulated Reconstructed EEG Sources", Internship Student (2021).

[14] **Diogo Sousa Morais**, "Estimating the effectiveness of source localized EEG for BCIs", Internship Student (2021).

[15] **Guilherme de Borras Silva**, "Cluster Permutation Analysis of N-Back related EEG-ERP Data", Internship Student (2021).

[16] **Iris Vantieghem**, "Using Neural Networks to derive Directed Connectivity between Reconstructed EEG Sources", Master of Science in Artificial Intelligence (2020 - 2021).

[17] **Didier Quintius**, "Neural Network Approach to the Inverse Problem", Master of Science in Artificial Intelligence (2020 - 2021).

Talks, presentations and other media

[1] **Federate What Transfers: Component-Aware Federation of Self-Supervised Backbones for Longitudinal 3D MRI**, "IEEE FLTA (International Conference on Federated Learning Technologies and Applications)", Oct. 2026.

[2] **Modality-Specific Encoding with a Shared Backbone for Early HCC Prediction from Multi-Modal Liver MRI**, "4th XAI-NL Meet-Up", 2026.

[3] **Optimizing Federated Block-Term Tensor Regression - Strategy Comparisons and Applications,** "Symposium on Federated Learning and Intelligent Computing Systems (FLICS)", 2025.

[4] **Federated Learning for Healthcare - From Brain-Computer Interfaces to Disease Prediction,** "IEEE FLTA (International Conference on Federated Learning Technologies and Applications)", 2025.

[5] **OHDSI Europe Symposium 2025,** "OHDSI Europe Symposium", 2025.

[6] **Guest Lecture "Federated Finger movement decoding - brain-computer interfacing",** "KULeuven", 2025.

[7] **Use Case - Real-World Evidence,** "Flanders Artificial Intelligence Research Program (FAIR) Research Day", 2024.

[8] **FAIR Use Case Real World Evidence Kick-off Event,** "Flanders Artificial Intelligence Research Program (FAIR)", 2024.

[9] **Decoding finger movements from invasive recordings in human motor cortex,** "Mindseed event Leuven, NeuroTech Leuven", 2023.

[10] **Guest Lecture "Finger movement decoding - brain-computer interfacing",** "KULeuven", 2023.

[11] **Coordinated Finger Movements Predicted from Intracranial Brain Activity,** "International Congress Humanities vs Sciences & the Knowledge Accelerating in Modern World: Parallels an Interaction,", 2022.

[12] **BCI demo - Advanced Engineering, Antwerp Expo,** "AI Flanders, Flanders Industry 4.0", 2022.

[13] **Guest Lecture "Decoding single and coordinated finger actions from intracranial brain activity",** "KULeuven", 2022.

[14] **Finger abduction trajectory prediction from high-density ECoG,** "Leuven AI Scientific Workshop", 2022.

[15] **Decoding single and coordinated finger actions from intracranial brain activity.,** "XIV World Scientific Congress - SCIENCE FOR PEACE Modern Science, Global and Regional Theory and Practice", 2021.

[16] **Guest Lecture "Finger Movement Decoding - From Source-Localisation to Tensor Regression Modelling",** "KULeuven", 2021.

[17] **Reconstructing single finger trajectories from intracranial brain activity,** "III International Conference "Volga Neuroscience Meeting 2021"", 2021.

[18] **Presentation "'MINDSPELLER' Medical Research Project on Brain Computer Interfaces" and concert (with Tigran Maytesian and his Mind Speller Chamber Orchestra),** "Cathedral of St. Michael and St. Gudula, Brussels", 2019.

[19] **Purpose-Centric Appropriation of Everyday Objects as Game Controllers,** "ACM SIGCHI", 2016.

References

Maryam Amir Haeri, m.amirhaeri@utwente.nl — *Associate Professor (Postdoc advisor)*
Liesbet M. Peeters, liesbet.peeters[at]uhasselt.be — *Associate Professor (Postdoc advisor) Biomedical Data Sciences*
Marc Van Hulle, marc.vanhulle[at]kuleuven.be — *Full Professor (PhD advisor) Laboratory for Neuro- and Psychophysiology*
Yves Moreau, yves.moreau[at]kuleuven.be — *Full Professor (Collaborator)*